

FOSSEE, IIT BOMBAY

eSim Semester Long Internship Autumn 2026

eSim PCB Design Project Report (Task 7)

ESP32-CAM Based TinyML Hand Classifier Carrier Board

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Repository: github.com/Aryakuna14/eSim_PCB_Design_Project_Files

EDA Tool: eSim 2.5

Internship: eSim Semester Long Internship Autumn 2026

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1. Project Overview

This project entails the design of a custom, dual-layer Printed Circuit Board (PCB) using the eSim (KiCad-based) environment to serve as a highly reliable hardware carrier for an ESP32-CAM module. The primary objective is to support a TinyML hand-sign classifier application at the edge. By migrating from a fragile breadboard prototype to a dedicated PCB, this design guarantees a stable, high-current power supply, eliminates electrical noise from loose jumper wires, and provides robust peripheral interfaces necessary for continuous machine learning inference and video streaming.

2. Circuit Description

The schematic architecture was developed entirely from scratch to address the specific power constraints of the ESP32-CAM module during heavy computational workloads.

Power Regulation

To prevent voltage drops during Wi-Fi transmission and camera operation, the board bypasses standard low-current supplies by integrating an AMS1117 3.3V linear regulator (U1). A 5V input from a terminal block is heavily filtered by 10 μ F and 1000 μ F capacitors before being stepped down to a clean 3.3V for the ESP32-CAM.

Peripherals & Control Logic

The circuit features a status LED (with a 330 Ω current-limiting resistor) and two hardware pushbuttons (SW1, SW2) configured with 10k Ω pull-up resistors. These buttons manage the hardware reset and pull GPIO0 low to enable serial flashing.

Expansion Interfaces

Multiple header connectors (J1 to J3) are integrated to provide accessible I2C expansion and direct external serial connections.

3. Component List (Bill of Materials)

Integrated Circuits & Modules

- **U1:** AMS1117-3.3 (3.3V Linear Voltage Regulator)
- **U2:** ESP32-CAM Module

Connectors & Headers

- **J1:** 2-Pin Header / Terminal Block (For 5V Power Input)
- **J2:** 4-Pin Header (Expansion / GPIO)
- **J3:** 4-Pin Header (Expansion / I2C / Serial)

Switches & Optoelectronics

- **SW1:** Pushbutton (Hardware Reset / EN pin)
- **SW2:** Pushbutton (Flash mode / GPIO0 pin)
- **D1:** Standard LED (Status Indicator)

Passive Components (Resistors)

- **R1, R2:** 10k Ω Resistors (Pull-up resistors for the pushbuttons)
- **R5:** 330 Ω Resistor (Current-limiting resistor for the status LED)
- **R3, R4:** 4.7k Ω Resistors (Likely I2C pull-ups for the expansion headers)

Passive Components (Capacitors)

- **C1, C2:** 10 μ F Capacitors (Input and output filtering for the AMS1117 regulator)
- **C3:** 100nF Capacitor (High-frequency decoupling)
- **C4:** 100 μ F or 1000 μ F Bulk Electrolytic Capacitor (To handle the massive current spikes when the ESP32 Wi-Fi and camera activate)

4. Schematic Screenshots

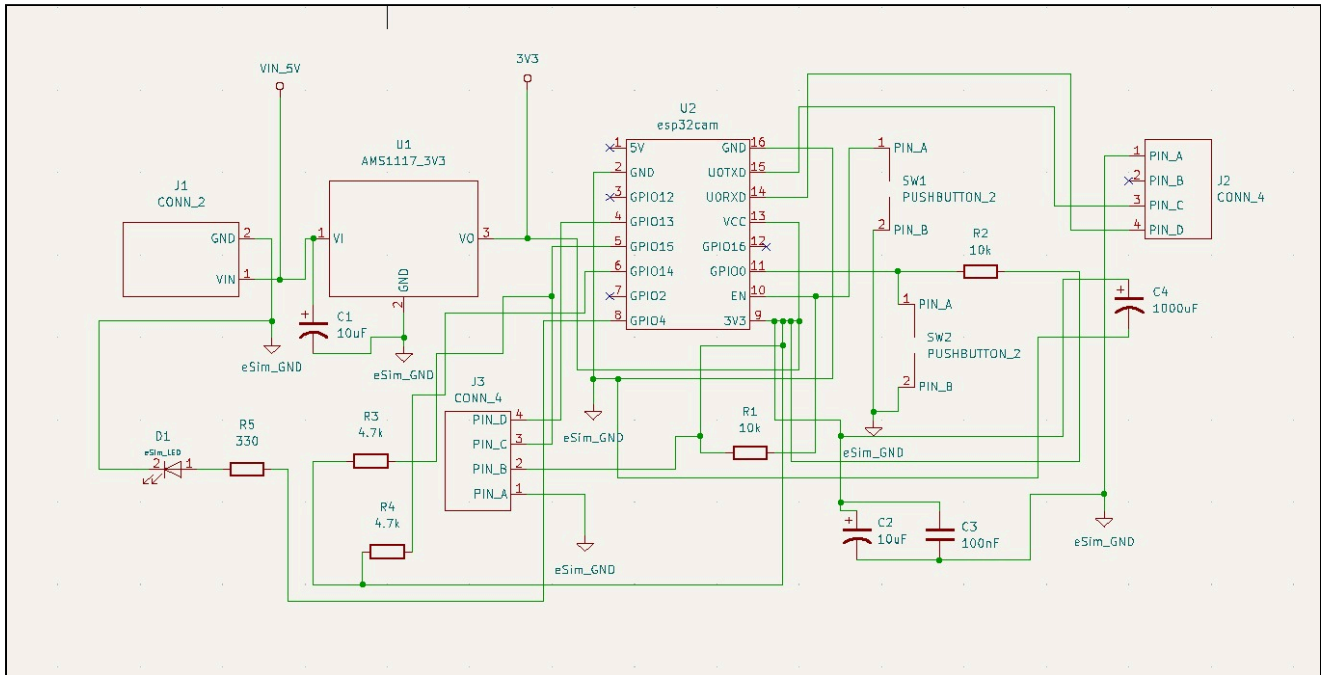


Figure 1: Complete schematic of the ESP32-CAM TinyML Hand Classifier carrier board

5. Footprint Assignment Details

The majority of the component footprints, including the central ESP32-CAM module, were custom-built to match specific physical dimensions.

Custom Library Implementation

A dedicated `esp32cam.pretty` library folder was created to house the custom ESP32-CAM footprint (U2).

Local Library Mapping

To ensure full compatibility when transferring the eSim workspace, these custom footprints were explicitly registered within the project's local `fp-lib-table` and `sym-lib-table`. This guarantees that the netlist generation and PCB updates compile flawlessly without footprint not found errors on the evaluator's machine.

6. PCB Layout Screenshots

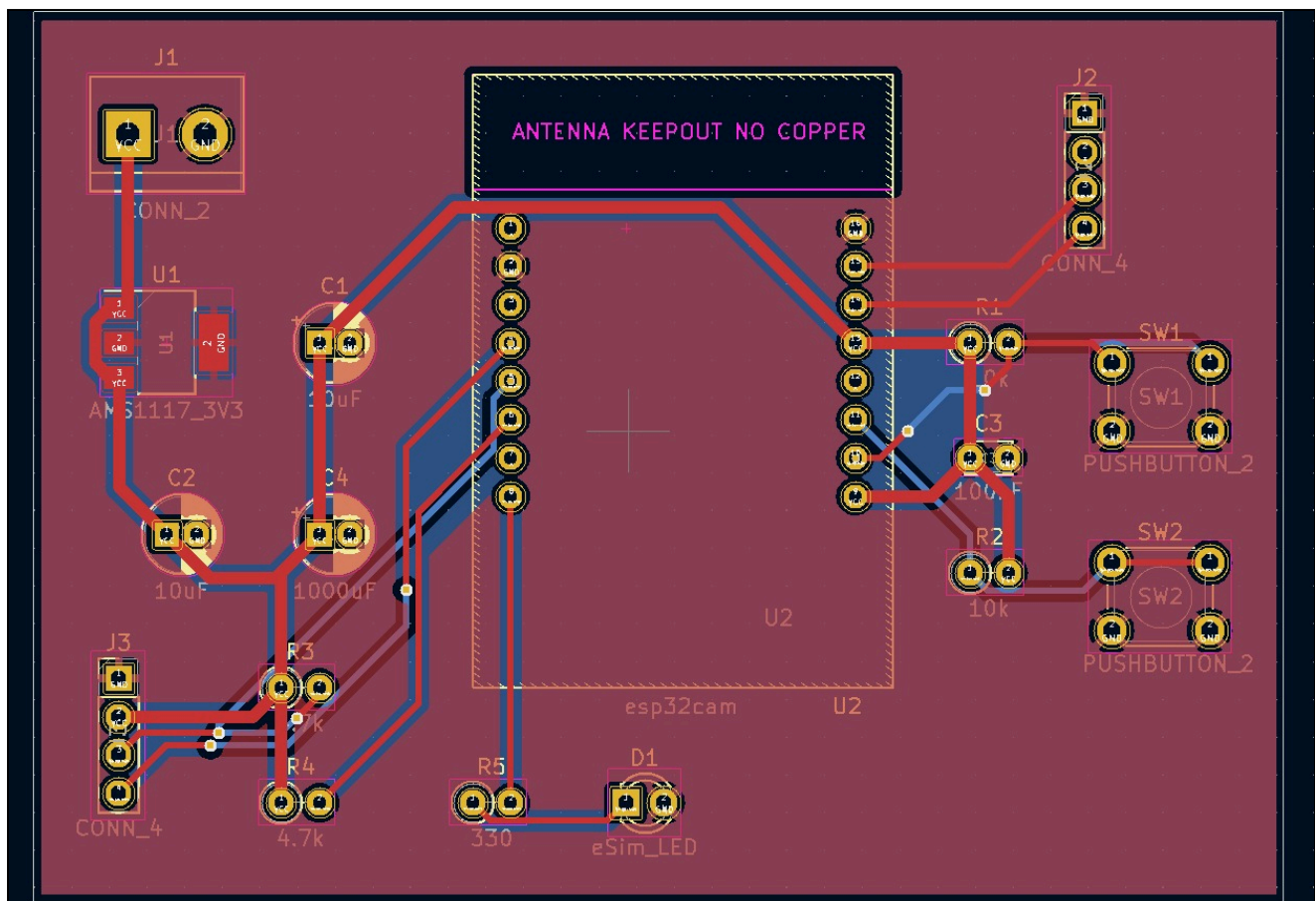


Figure 2: 2D PCB layout showing front copper (F.Cu) and back copper (B.Cu) routing

7. Routing Methodology

The board was routed using a two-layer methodology, strictly prioritizing power integrity and signal clarity.

Trace Width Strategy

A dual-width routing scheme was enforced. High-current power delivery networks (5V and 3.3V) were routed using thick **0.8 mm** traces to minimize voltage drop and handle peak current demands. Standard data and logic signals were routed using **0.4 mm** traces.

Layer Transitioning

Signal trapping was actively avoided by utilizing orthogonal routing practices between the top and bottom copper layers, connected via standard vias.

Trace Necking

Where the 0.8 mm power traces intersected tight component pad clearances, a trace necking technique was utilized to safely transition the track width, preventing clearance violations.

8. DRC Results

The final board successfully passes the automated Design Rules Check (DRC) with **0 Errors**, verifying electrical validity and manufacturability.

DRC Result: 0 Errors The board passes all design rule checks and is electrically valid for fabrication.

The report lists 15 cosmetic/non-blocking warnings, which were intentionally left as they do not impact functionality:

- **Via not connected on only one layer:** Applies to redundant single-layer tie points (e.g. R1-Pad2, J3-Pad4).
- **Silkscreen clipped by solder mask (Rect on Margin):** These are overlapping margin-layer rectangles leftover from the custom footprint creation process. The fabrication export correctly ignores these without affecting the physical solder mask.

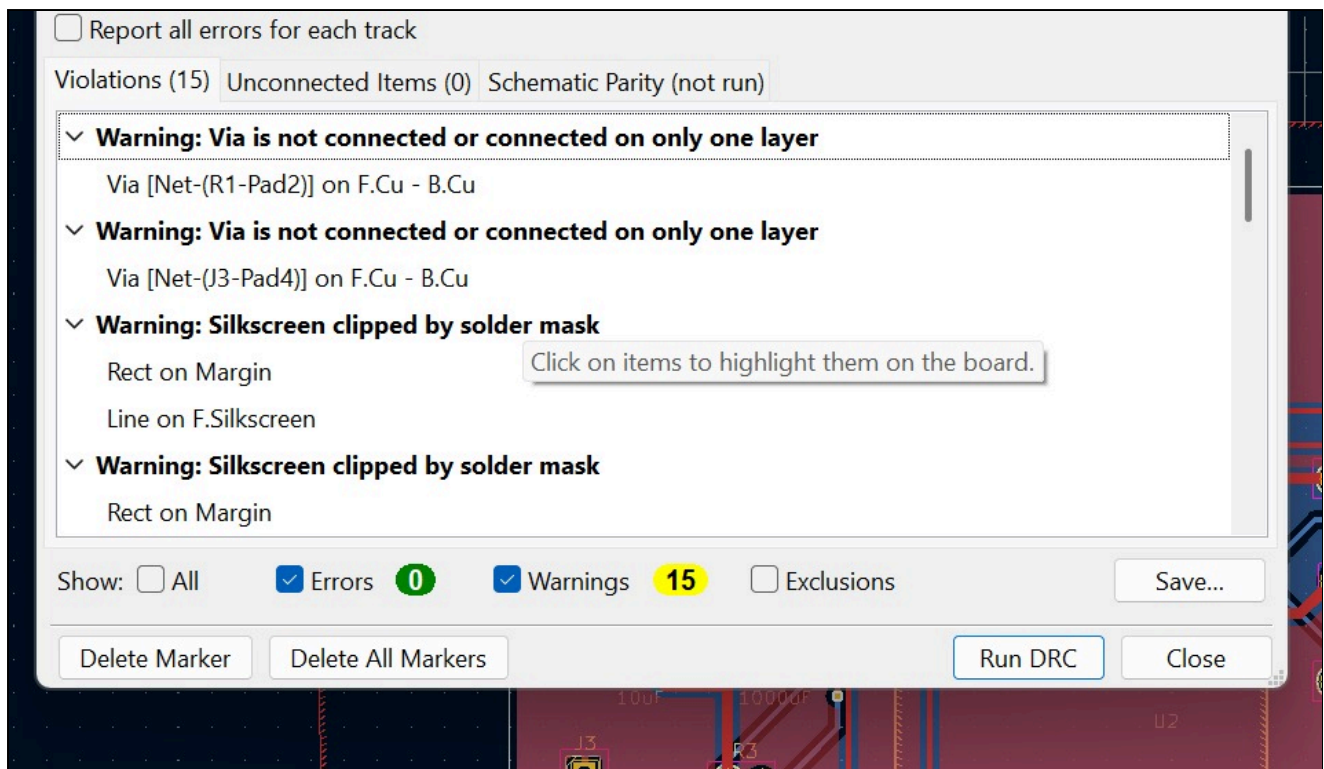


Figure 3: eSim DRC results window confirming 0 Errors

9. 3D View Screenshots

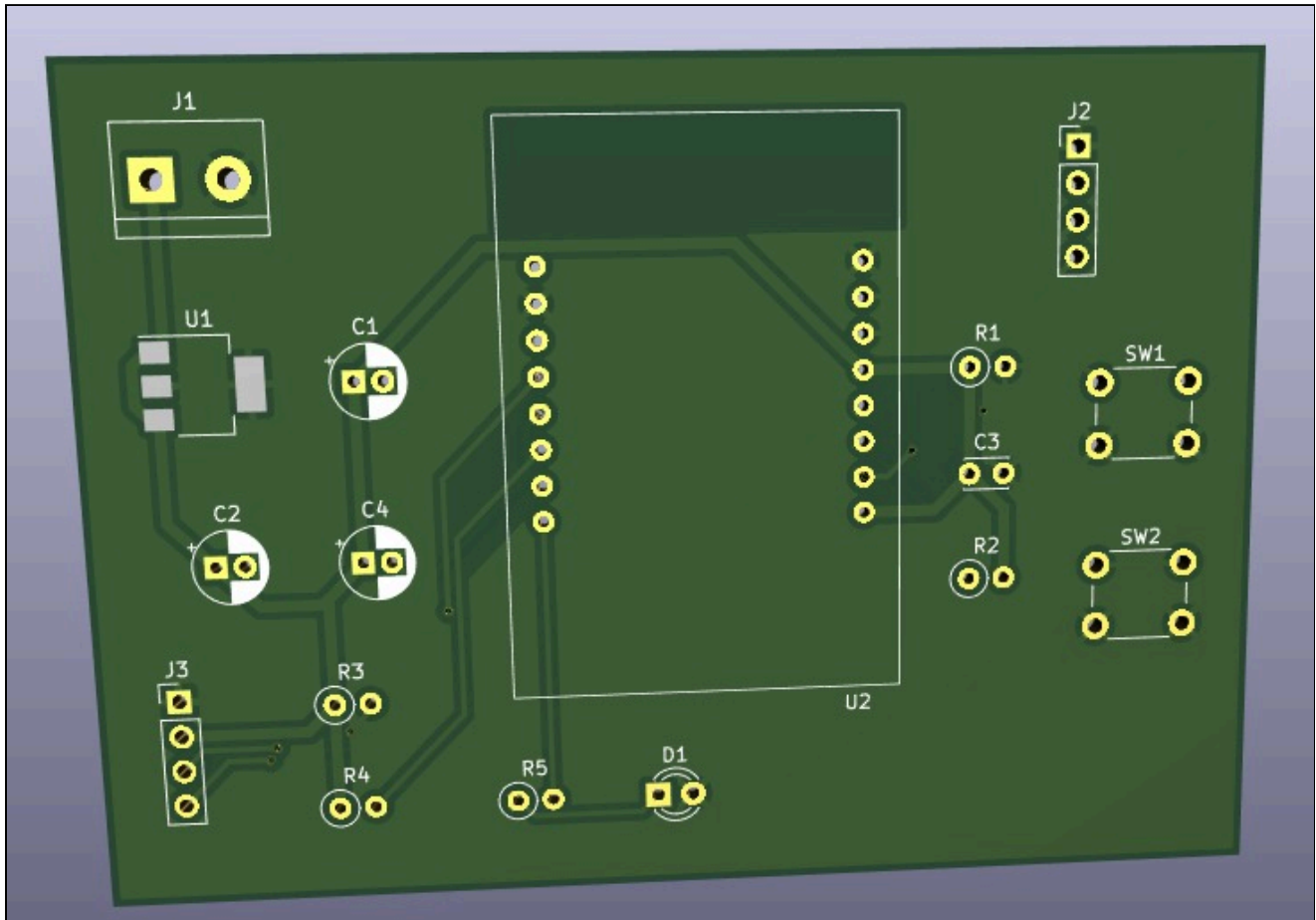


Figure 4: 3D rendered view of the assembled ESP32-CAM TinyML Hand Classifier PCB

10. Design Considerations and Conclusions

Critical Design Consideration: Antenna RF Keepout Zone

A major design requirement was maintaining the RF integrity of the ESP32-CAM's onboard Wi-Fi antenna. A specific Rule Area (labeled **ANTENNA KEEPOUT: NO COPPER**) was placed over the top strip of the U2 footprint. During development, a critical EDA behavior was identified: creating a Rule Area without explicitly selecting the target layers (F.Cu/B.Cu) causes the software to silently enforce keepout constraints globally across the entire board, resulting in blocked routing. Diagnosing this and isolating the rule strictly to the physical antenna footprint successfully cleared the interference and protected the RF zone.

Conclusion

This custom PCB successfully fulfills all technical parameters of Task 7. By utilizing custom footprint libraries, adhering to strict high-current power routing principles, and thoroughly debugging DRC constraints to achieve a 0-error board, this project provides a highly robust hardware foundation for executing edge-based machine learning models.

Repository: github.com/Aryakuna14/eSim_PCB_Design_Project_Files

Pull Request: [FOSSEE/eSim_PCB_Design_Project_Files #2](#)